

Project Report: Clustering with Neural Networks using Hugging Face Datasets

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1 Introduction

This report evaluates the implementation of a neural network model for clustering using the `gimmaru/glue-sst2` dataset from Hugging Face[cite: 5, 6]. The project focuses on developing an autoencoder-based neural network to encode text data into a low-dimensional latent space, followed by K-Means clustering to group sentences by sentiment[cite: 7]. The implementation is assessed against requirements focusing on dataset handling, architecture, training strategy, and evaluation[cite: 8].

2 Objective

The primary objective was to design and implement a neural network capable of performing clustering on the specified dataset[cite: 10, 11]. Performance is evaluated using metrics such as Silhouette Score, Davies-Bouldin Index, and clustering accuracy, with visualization via t-SNE[cite: 11].

3 Implementation Overview

The pipeline processes text from `gimmaru/glue-sst2`, extracts BERT embeddings, trains an autoencoder to learn a 64-dimensional latent space, and applies K-Means clustering[cite: 13, 14].

3.1 Software and Tools

The project utilizes the following environment [cite: 16-22]:

- **Language:** Python 3.8+ (Assumed 3.11 in Google Colab)[cite: 17, 24].
- **Frameworks:** PyTorch, Scikit-learn, and Hugging Face Transformers[cite: 18, 19, 20].
- **Hardware Support:** Supports CUDA-enabled GPUs via `torch.device`[cite: 25].

3.2 Dataset Handling and Preprocessing

The implementation uses `bert-base-uncased` for text embeddings.

- **Embeddings:** Extracts 768-dimensional CLS token embeddings.
- **Tokenization:** Uses `SST2Dataset` class with a `max_length` of 128[cite: 69].

4 Neural Network Design

The architecture follows an autoencoder design with a 64-dimensional latent space[cite: 73, 78].

4.1 Architecture Details

- **Encoder:** `nn.Linear(768, 256) → ReLU → nn.Linear(256, 64) → ReLU`[cite: 78].
- **Decoder:** `nn.Linear(64, 256) → ReLU → nn.Linear(256, 768) → Tanh`[cite: 79].

4.2 Loss Function

The project outlines a combined loss requirement[cite: 74, 75]:

$$L = L_{\text{reconstruction}} + \lambda \cdot L_{\text{clustering}} \quad (1)$$

Note: The current implementation utilizes MSE reconstruction loss but applies K-Means post-training rather than as a joint loss[cite: 80, 81].

5 Challenges and Solutions

5.1 Dataset Loading Error

A `NotImplementedError` was encountered due to Colab’s filesystem restrictions and the Parquet-based loading script of `gimmaru/glue-sst2`[cite: 122, 123].

Solution: The recommended fix involves clearing the Hugging Face cache and explicitly defining a custom cache directory[cite: 129, 130, 131]:

```
import os
import shutil
cache_dir = os.path.expanduser("/tmp/hf_datasets")
# Code to clear and reload with verification bypass
dataset = load_dataset("gimmaru/glue-sst2", split="validation",
                      cache_dir="/tmp/hf_datasets",
                      ignore_verifications=True)
```

6 Evaluation and Compliance

The project demonstrates high compliance with the original outline[cite: 153].

Metric	Status
Silhouette Score	Implemented [cite: 104]
Davies-Bouldin Index	Implemented [cite: 104]
Clustering Accuracy	Implemented [cite: 104]
Calinski-Harabasz Index	Missing (Minor deviation) [cite: 106]
t-SNE Visualization	Implemented [cite: 111]

Table 1: Implementation Compliance Summary

7 Conclusion

The implementation effectively meets the core project requirements, providing a robust pipeline for BERT embeddings and autoencoder-based clustering[cite: 176, 177]. With the recommended dataset loading fixes, the model serves as a functional tool for sentiment-based sentence clustering[cite: 178, 180].